

RESTRICTED DYNAMIC GEOMETRIC COMPLEXITY: PATH-SPACE REDUCTION AND MÖBIUS–JACOBI RESPONSE

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Abstract. Structured preconditioners restrict optimization to a small family of positive metrics, but endpoint condition-number reachability does not measure the geometric effort required to reach a useful metric. We formulate this effort as a path-space value problem. Restricted dynamic geometric complexity is the least affine-invariant length of an admissible metric path whose endpoint reaches a Hessian-relative generalized-eigenvalue condition target. Path elimination gives an exact min-plus semigroup and Bellman principle, while fixed-horizon kinetic energy is exactly squared complexity divided by twice the horizon. Explicit Tonelli and compact-control realizations supply existence and a conditional Hamilton–Jacobi layer. The main response result is global on a Hadamard state space: geodesic convexity produces a smooth intervention-cube path branch and a uniformly coercive Jacobi form, while one Green inverse generates the value Hessian, two-sided force-to-curvature bounds, exact Möbius effects, and arbitrary prescribed finite-order responses. For the hard condition target, a bordered Jacobi–KKT theorem differentiates the moving projection endpoint and multiplier on every regular active spectral stratum; its indefinite inverse also explains why hard-target interactions need not share the unconstrained sign. The theory specializes to affine-invariant positive-definite geometry. A determinant-one two-dimensional diagonal model has an exact target interval, a closed-form forced path, and a strictly negative-definite interaction matrix. A moving diagonal Hessian gives a closed-form hard-target projection, multiplier, and pair effects of either sign, while a coordinate-sequential three-dimensional protocol yields an exact path metric strictly larger than the ambient projection distance. Thus the global Green and bordered hard-target responses are explicit laws of restricted metric-path elimination built on Bellman composition.

Key words. structured preconditioning, affine-invariant geometry, dynamic programming, Jacobi operator, parametric optimization, Möbius interaction

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1. Introduction. Structured preconditioners constrain optimization to a small family of positive metrics. This creates an endpoint question and a path question. The endpoint question asks whether the family can reach a prescribed condition number. The path question asks how much intrinsic metric motion is required to reach it from the current state. Endpoint feasibility alone does not measure this geometric effort: two families may reach the same target but place it at very different affine-invariant distances from the initial metric.

We formulate the path question as a value problem. For a quadratic objective with Hessian $H \succ 0$, a metric $G \succ 0$ induces the generalized eigenproblem $Hv = \lambda Gv$. In relative coordinates

$$S = H^{-1/2}GH^{-1/2},$$

the relevant condition ratio is

$$\kappa_{\text{gen}}(H, G) := \frac{\lambda_{\max}(G^{-1/2}HG^{-1/2})}{\lambda_{\min}(G^{-1/2}HG^{-1/2})} = \kappa(S).$$

This is the positive generalized-eigenvalue ratio of $Hv = \lambda Gv$, which is invariant under inversion of the relative spectrum. Given a metric family $\mathfrak{F}$, an initial state S_0 , and $K \geq 1$, restricted dynamic geometric complexity $D_{K, \mathfrak{F}}(S_0; H)$ is the least affine-invariant length of an admissible metric path that reaches

$$\mathcal{C}_K = \{S \in \mathbb{S}_{++}^d : \kappa(S) \leq K\}.$$

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The value is $+\infty$ when the target is structurally unreachable. The word complexity refers throughout to minimum intrinsic metric motion (or, after fixing the horizon, its exactly equivalent action value); it carries no iteration-count or wall-clock complexity claim.

The central operation is elimination of an entire future path. If $c_{s,t}(x, y)$ denotes the least action between two states, then restriction and concatenation yield

$$c_{s,t}(x, z) = \inf_y \{c_{s,r}(x, y) + c_{r,t}(y, z)\}.$$

This min-plus semigroup is the exact global law of path elimination. It precedes smoothness, existence, and Hamilton–Jacobi regularity. Tonelli conditions supply attained paths, while a separate compact-control realization supplies the HJB characterization. For fixed-horizon kinetic action, constant-speed gauge fixing gives the exact identity

$$\inf_{\gamma} \frac{1}{2} \int_0^T \|\dot{\gamma}_t\|^2 dt = \frac{D_{K,\mathfrak{F}}(S_0; H)^2}{2T}.$$

Thus the smooth action used for response analysis has precisely the same geometric minimizers as length RDGC.

Path elimination also has a differential law. Near a unique smooth optimum, the vertical second variation is a Jacobi operator $\mathcal{J}_{\text{Jac}}$. Eliminating path fluctuations gives

$$D\gamma_{\star} = -\mathcal{J}_{\text{Jac}}^{-1} \mathcal{A}_{\gamma z}, \quad D^2 \text{Val} = \mathcal{A}_{zz} - \mathcal{A}_{z\gamma} \mathcal{J}_{\text{Jac}}^{-1} \mathcal{A}_{\gamma z}.$$

The second term is the path-space Schur complement. For additive interventions it is a negative-semidefinite Gram operator. Its mixed entries integrate over cube faces to exact Boolean Möbius effects.

The local formula becomes global under geometric hypotheses. On a Hadamard state space, smooth geodesically convex bulk and terminal potentials yield a unique optimal path for every intervention in a neighborhood of the Boolean cube. Nonpositive curvature makes the full Jacobi index form uniformly coercive. The Green inverse then solves bulk forcing and terminal Robin forcing simultaneously, generates pair and conditional Möbius effects, and recursively determines every prescribed finite interaction order under the corresponding smoothness. Its Gram representation also identifies the exact nullspace and rank of value curvature and bounds that curvature by the intervention-force norm and the Jacobi spectrum.

The hard condition target has a different local operator. On a regular active spectral stratum, terminal feasibility adds one multiplier and one linearized boundary equation. The resulting bordered Jacobi–KKT inverse jointly differentiates the optimal path, moving projection endpoint, and multiplier, and yields the constrained value Hessian. This directly treats a moving hard RDGC target while locating the precise failure boundary at eigenvalue collisions, loss of strict complementarity, or active-stratum changes. A determinant-one diagonal RDGC model realizes the unconstrained Green theory in closed form and produces a strictly negative-definite interaction matrix.

Contributions.. The paper establishes four results.

1. We define RDGC as an affine-invariant path distance to a Hessian-relative generalized spectral target and derive the full-SPD spectral benchmark. Exact min-plus composition and the length–energy identity provide its path-elimination and fixed-horizon foundations.

2. We prove a global Hadamard Möbius–Jacobi theorem. One uniformly coercive Green operator handles bulk and terminal forcing, gives exact force-to-curvature bounds and rank identities, integrates to pair and conditional effects, and recursively generates every prescribed finite interaction order.
3. We prove a bordered Jacobi–KKT theorem for a genuinely hard terminal inequality. On a smooth active condition-number stratum it differentiates the moving RDGC projection and multiplier and gives an exact constrained Möbius formula whose sign can differ from the unconstrained Gram law.
4. We give explicit Tonelli and compact-control realizations, specialize the response theory to affine-invariant SPD geometry, and solve a two-dimensional determinant-one diagonal RDGC model exactly. Both the fixed-endpoint Green response and a moving hard condition target have closed-form paths and interactions; the latter realizes both interaction signs. A three-dimensional coordinate-sequential update protocol has an exact ℓ_1 -type RDGC strictly larger than its ambient AIRM projection, showing where the declared path class changes the optimization value. The finite-dimensional companion [30] proves generic Schur response and affine-intervention Gram formulas for hidden controller states. Those formulas are the finite-dimensional template here. The present paper’s new response results begin with their path-Hilbert realization: Hadamard geometry derives global existence and cube-wide coercivity, the weak inverse becomes a strong Green–Jacobi problem with Robin forcing, arbitrary prescribed finite orders share that inverse, and a bordered boundary system treats the moving hard condition target. The companion information-induced-geometry paper studies exact visible metric completion [29]. The present work is self-contained at the path-space level and does not depend on the subsequent causal optimizer calculus.

Organization.. Section 2 positions the results relative to optimal control, parametric optimization, SPD geometry, and interaction decompositions. Section 3 defines the affine-invariant optimization target. Section 4 develops path elimination and global response. Section 5 gives the exact diagonal realization, and sections 6 and 7 record verification and scope.

2. Related Work.

Variational optimization and optimal control.. Continuous-time optimization has been studied through gradient flows, accelerated ODEs, Bregman Lagrangians, mirror descent, and natural-gradient geometry [40, 43, 25, 6, 2]. Our controlled variable is the metric itself, the terminal set is determined by preconditioning quality, and the value measures the intrinsic deformation required to reach that set.

Min-plus semigroups, dynamic programming, and Hamilton–Jacobi equations are classical in optimal control [17]. Geometric control also supplies the classical second variation, Jacobi fields, and conjugate point analysis [1]. We use these ingredients as the analytic foundation for a different reduction target: an entire metric path ending at a preconditioning-quality set. The new response statements are the intervention-cube Green representation and its bordered hard-target analogue.

Parametric optimization and Jacobi response.. Reduced Hessians, variable projection, solution-map sensitivity, and implicit differentiation eliminate optimized variables through Schur complements [19, 9, 15, 3, 8, 18]. We do not claim the generic finite-dimensional Schur formula as new. The contribution is its coercive path-Hilbert realization, the derivation of that coercivity from Hadamard geometry on an entire intervention neighborhood, the joint bulk/terminal Green operator, and the resulting finite- and arbitrary-order interaction formulas. For an active hard terminal inequality,

the closest general framework is parametric constrained optimization and generalized equations [9, 15]; our bordered theorem identifies the derivative explicitly as a Jacobi–KKT boundary operator and specializes it to the moving condition-number projection.

Jacobi fields and index forms classically describe variations of geodesics. Here the Jacobi operator contains kinetic, curvature, bulk-potential, and terminal Robin terms. Nonpositive curvature contributes a nonnegative term to the index form, allowing global control even when the convex potentials have linear negative growth.

The finite-dimensional companion theory already contains the generic hidden response, Schur reduced Hessian, affine-intervention negative Gram operator, and its pair finite-difference integral [30]. The local formulas in Theorem 4.9 and Corollary 4.15 are their Hilbert-path lift and are used here as infrastructure. The additions proved in this paper are the Hadamard hypotheses that derive global path existence, uniqueness, and cube-wide coercivity; the strong Green–Jacobi boundary problem that unifies bulk and Robin forcing; arbitrary prescribed finite-order path recursion; and the bordered Jacobi–KKT response of a moving hard target. These conclusions are absent from the finite-dimensional companion reduction theorem.

Positive-definite geometry and structured preconditioning. The affine-invariant geometry of positive-definite matrices supplies the distance, exponential map, and nonpositive curvature used in the SPD specialization [7, 35, 22]. Geodesic convex analysis and metric projection on Hadamard spaces are developed in [39, 31, 5]. Smoothness on a fixed simple-eigenvalue stratum and loss of classical differentiability at spectral multiplicities follow the spectral-function theory in [26, 28, 27].

Classical equilibration and condition-number minimization optimize an endpoint scaling or matrix approximation [42, 33, 32, 13, 41, 37, 14]. These works clarify the endpoint benchmark addressed by Theorem 3.1. RDGC adds the initial metric and an admissible path geometry; its hard-target theorem differentiates the projection path in addition to the endpoint condition number.

Diagonal, block, and Kronecker preconditioners arise in adaptive and second-order optimization [16, 24, 38, 34, 20, 21, 4]. These algorithms motivate restricted metric families, but a concrete optimizer-to-path statement must additionally specify the optimizer state, damping, bias correction, step map, and interpolation. RDGC is conditional on that realization and measures the geometric effort once it has been declared.

Finite interaction decompositions. Boolean and factorial interactions can be represented by finite differences, Möbius coefficients, and higher-order attribution indices [36, 11, 12, 23]. The result here is a path-geometric representation: the pair coefficient is a cube-face integral of an inverse-Jacobi Gram kernel, conditional effects use the same kernel on subfaces, and all higher coefficients follow from one Green recursion.

3. Restricted Metric Paths in Affine-Invariant Geometry. We now specialize the path system to positive-definite metrics and make the terminal optimization problem explicit. Let

$$f(\theta) = \frac{1}{2} \theta^\top H \theta, \quad H \in \mathbb{S}_{++}^d,$$

and let $G \in \mathbb{S}_{++}^d$ be a positive metric. In Hessian-relative coordinates,

$$S = H^{-1/2} G H^{-1/2}.$$

184 The Hessian-relative condition ratio is

$$185 \quad \kappa_{\text{gen}}(H, G) := \frac{\lambda_{\max}(G^{-1/2}HG^{-1/2})}{\lambda_{\min}(G^{-1/2}HG^{-1/2})} = \kappa(S).$$

186 Whenever the shorthand $\kappa(G^{-1}H)$ appears, it denotes this ratio of positive generalized
187 eigenvalues, never the Euclidean singular-value condition number of the generally
188 nonnormal product $G^{-1}H$. The affine-invariant metric and distance are

$$189 \quad \|Z\|_{S, \text{AI}} = \|S^{-1/2}ZS^{-1/2}\|_F, \quad d_{\text{AI}}(S_0, S_1) = \|\log(S_0^{-1/2}S_1S_0^{-1/2})\|_F.$$

190 The cone $\mathbb{S}_{++}^d$ is a finite-dimensional Hadamard manifold [7, 35].

191 For $K \geq 1$, define the condition target

$$192 \quad \mathcal{C}_K = \{S \in \mathbb{S}_{++}^d : \kappa(S) \leq K\}.$$

193 The unrestricted distance to this target has a one-dimensional spectral formula.

194 THEOREM 3.1 (Full-SPD spectral benchmark). *Let $y_1 \leq \dots \leq y_d$ be the ordered*
195 *log-eigenvalues of $S_0 \in \mathbb{S}_{++}^d$. Then*

$$196 \quad D_K(S_0) := d_{\text{AI}}(S_0, \mathcal{C}_K) = \min_{c \in \mathbb{R}} \left(\sum_{i=1}^d \text{dist}(y_i, [c, c + \log K])^2 \right)^{1/2}.$$

197 Let $\mathfrak{F} \subset \mathbb{S}_{++}^d$ be an admissible family and set

$$198 \quad \mathcal{S}_{\mathfrak{F}}(H) = \{H^{-1/2}GH^{-1/2} : G \in \mathfrak{F}\}.$$

199 A realization includes a path-connected component, a treatment of scale freedom, an
200 admissible absolutely continuous path class, and its induced or quotient affine-invariant
201 length.

202 DEFINITION 3.2 (Restricted dynamic geometric complexity). *For $G_0 \in \mathfrak{F}$ and*
203 *$S_0 = H^{-1/2}G_0H^{-1/2}$, define*

$$204 \quad D_{K, \mathfrak{F}}(S_0; H) = \inf \{ \text{Len}_{\text{AI}}(S_{[0, T]}) : S(0) = S_0, S(T) \in \mathcal{C}_K, \\ S_t \in \mathcal{S}_{\mathfrak{F}}(H), S_{[0, T]} \text{ admissible} \}.$$

205 *The value is $+\infty$ if the relative target is empty or belongs to a different admissible*
206 *component.*

207 For a closed geodesically convex realization, RDGC is an ordinary Hadamard
208 projection distance. This fact connects the nonsmooth length problem to the fixed-
209 horizon action used in the response theory.

210 PROPOSITION 3.3 (Static RDGC projection). *Suppose that $\mathcal{S}_{\mathfrak{F}}(H)$ is nonempty,*
211 *closed, and geodesically convex and that*

$$212 \quad \mathcal{T}_{K, \mathfrak{F}}(H) = \mathcal{S}_{\mathfrak{F}}(H) \cap \mathcal{C}_K$$

213 *is nonempty. Then S_0 has a unique projection $P_{K, \mathfrak{F}}(S_0; H)$ onto $\mathcal{T}_{K, \mathfrak{F}}(H)$, and*

$$214 \quad D_{K, \mathfrak{F}}(S_0; H) = d_{\text{AI}}(S_0, P_{K, \mathfrak{F}}(S_0; H)).$$

215 For quadratic kinetic action with terminal indicator of $\mathcal{T}_{K,\mathfrak{F}}(H)$, the value at time
 216 $t < T$ is

$$217 \quad V(t, S) = \frac{d_{\text{AI}}(S, \mathcal{T}_{K,\mathfrak{F}}(H))^2}{2(T-t)}.$$

218 The minimizing path is the unique constant-speed geodesic to the projection.

219 The projection reduction is a benchmark for unrestricted curves in a closed
 220 geodesically convex realization. It does not apply when the update protocol restricts
 221 admissible velocities or excludes the ambient geodesic. The exact coordinate-sequential
 222 model in [Proposition 5.3](#) gives a structured example whose RDGC strictly exceeds
 223 this ambient benchmark.

224 The convex potentials required by the global response theorem arise naturally
 225 from relative spectra. For $H_a \in \mathbb{S}_{++}^d$ and a convex permutation-invariant function
 226 $\phi_a : \mathbb{R}^d \rightarrow \mathbb{R}$, write

$$227 \quad \Omega_a(G) = \phi_a\left(\log \lambda(G^{-1/2} H_a G^{-1/2})\right).$$

228 PROPOSITION 3.4 (Geodesically convex spectral information). Let $\mathfrak{F} \subset \mathbb{S}_{++}^d$ be
 229 nonempty, closed, and geodesically convex. If $\rho : \mathbb{R}^q \rightarrow \mathbb{R}$ is convex and nondecreasing
 230 in every coordinate, then

$$231 \quad \mathcal{J}(G) = \rho(\Omega_1(G), \dots, \Omega_q(G))$$

232 is closed and geodesically convex on $\mathfrak{F}$. If every $\phi_a(x+c\mathbf{1}) = \phi_a(x)$, then $\mathcal{J}(cG) = \mathcal{J}(G)$.
 233 In particular,

$$234 \quad \phi_{\text{osc}}(x) = \max_i x_i - \min_i x_i$$

235 generates $\mathcal{J}_H(G) = \log \kappa_{\text{gen}}(H, G)$. Smooth spectral approximations retain geodesic
 236 convexity.

237 Combining this construction with the global path theorem gives the exact SPD
 238 interaction law used below.

239 COROLLARY 3.5 (SPD–AIRM Möbius–Jacobi specialization). Let $\mathcal{N}$ be $\mathbb{S}_{++}^d$, or
 240 a structured AIRM realization that is a complete simply connected totally geodesic
 241 manifold. Let the bulk and terminal potentials in [Theorem 4.11](#) be smooth geodesically
 242 convex spectral information potentials satisfying its regularity assumptions. Then the
 243 conclusions of that theorem hold with

$$244 \quad J_u \xi = -D_t^2 \xi - R^{\text{AI}}(\xi, \dot{G}_u) \dot{G}_u + \text{Hess}^{\text{AI}} W_u \xi,$$

245 and

$$246 \quad -\left\langle R^{\text{AI}}(\xi, \dot{G}_u) \dot{G}_u, \xi \right\rangle_{G_u} \geq 0.$$

247 In particular, the pair effect is

$$248 \quad \boxed{\zeta_{\{i,j\}} = - \int_{[0,1]^2} \langle D\Delta_i, \mathfrak{I}_u^{-1} D\Delta_j \rangle du_i du_j.}$$

249 Fixed smooth active strata are covered. The nonsmooth condition-number width requires
 250 a generalized Jacobi inclusion at eigenvalue or active-index changes.

Remark 3.6 (Structured families covered directly). The positive diagonal cone in a fixed basis and every fixed block-diagonal SPD cone are complete simply connected totally geodesic AIRM submanifolds; their geodesics, logarithms, and exponentials preserve the declared block structure. The determinant-one diagonal family is the corresponding flat totally geodesic scale section. Hence the corollaries above apply directly to these families. Generic low-rank and Kronecker parameterizations require their own embedded or quotient geometry and are outside this direct totally-geodesic specialization.

The hard condition target itself also has a classical response away from spectral collisions. It is useful here to write

$$\kappa_{\text{gen}}(H, G) = \frac{\lambda_{\max}(G^{-1/2}HG^{-1/2})}{\lambda_{\min}(G^{-1/2}HG^{-1/2})}.$$

COROLLARY 3.7 (Moving hard-condition target on a smooth spectral stratum). *Let $\mathfrak{F}$ be a complete simply connected totally geodesic AIRM realization, fix $G_0 \in \mathfrak{F}$, and let $H_u \in \mathbb{S}_{++}^d$ and $K_u > 1$ vary smoothly. Define*

$$c_u(G) = \log \kappa_{\text{gen}}(H_u, G) - \log K_u, \quad \mathcal{T}_u = \{G \in \mathfrak{F} : c_u(G) \leq 0\}.$$

Suppose that $G_0 \notin \mathcal{T}_{u_0}$, that the largest and smallest generalized eigenvalues of (H_{u_0}, q_0) are simple at the AIRM projection $q_0 = P_{\mathcal{T}_{u_0}} G_0$, and that $d(c_{u_0}|_{\mathfrak{F}})(q_0) \neq 0$. Then, locally in u , the projection endpoint q_u , its constant-speed geodesic γ_u , and its positive terminal multiplier are smooth. Moreover,

$$\mathcal{V}_{\text{hard}}(u) = \frac{d_{\text{AI}}(G_0, \mathcal{T}_u)^2}{2T}$$

satisfies (HT4)–(HT5) with $\mathfrak{J}_u$ equal to the kinetic index form plus $\lambda_u \text{Hess}^{\text{AI}}(c_u|_{\mathfrak{F}})$ at the endpoint. If the active defining function is affine in the intervention coordinates on a Boolean face, and the same active spectral stratum, simple extreme eigenvalues, LICQ, positive multiplier, and bordered invertibility persist on that entire face, (HT6) gives its exact hard-RDGC interaction.

Thus the response differentiates the moving hard-target projection itself. The conclusion ends precisely when an extreme generalized eigenvalue loses simplicity, the restricted boundary loses regularity, strict complementarity fails, or the active stratum changes.

4. Path-Space Reduction and Jacobi Response. This section separates the algebraic, variational, and differential layers of restricted geometric control. The first layer needs only path concatenation. The second adds compactness and convexity. The third studies a smooth nondegenerate optimal path through its second variation.

4.1. Transition costs and min-plus composition. Let $\mathcal{M}$ be a state space and let $\mathcal{P}_{s,t}(x, y)$ be a declared class of admissible paths from x at time s to y at time t . A path has action $\mathcal{A}_{s,t}(\gamma) \in \mathbb{R} \cup \{+\infty\}$. Empty path classes have value $+\infty$.

ASSUMPTION 4.1 (Composable path system). *For every $s < r < t$:*

1. *restricting $\gamma \in \mathcal{P}_{s,t}(x, z)$ at time r produces paths in $\mathcal{P}_{s,r}(x, \gamma(r))$ and $\mathcal{P}_{r,t}(\gamma(r), z)$;*
2. *any two admissible segments with a common state at time r can be concatenated into an admissible path;*
3. *the action is additive under restriction and concatenation.*

4. for every fixed interval $[a, b]$, there is $m_{a,b} > -\infty$ such that $\mathcal{A}_{a,b}(\gamma) \geq m_{a,b}$ for every admissible segment.

Define the transition cost

$$c_{s,t}(x, y) = \inf_{\gamma \in \mathcal{P}_{s,t}(x, y)} \mathcal{A}_{s,t}(\gamma).$$

THEOREM 4.2 (Path-elimination semigroup). Under Theorem 4.1, for every $s < r < t$,

$$(P1) \quad c_{s,t}(x, z) = \inf_{y \in \mathcal{M}} \{c_{s,r}(x, y) + c_{r,t}(y, z)\}.$$

Both sides lie in $\mathbb{R} \cup \{+\infty\}$, so the min-plus sum is always defined. The equality does not require minimizers, smoothness, or convexity.

For a terminal cost $\Gamma : \mathcal{M} \rightarrow \mathbb{R} \cup \{+\infty\}$ bounded below, define

$$V(s, x) = \inf_y \{c_{s,T}(x, y) + \Gamma(y)\}.$$

COROLLARY 4.3 (Bellman principle). Under the same assumptions,

$$(P2) \quad V(s, x) = \inf_y \{c_{s,r}(x, y) + V(r, y)\}, \quad s < r < T.$$

Thus Bellman recursion is associativity of path elimination across time slices.

4.2. The fixed-horizon energy gauge. Let $\mathcal{C} \subset \mathcal{M}$ be a terminal set and let $\mathcal{P}_T(x, \mathcal{C})$ be an admissible class of absolutely continuous paths on $[0, T]$ starting at x and ending in $\mathcal{C}$. Suppose the class is closed under orientation-preserving absolutely continuous reparameterization and every finite-length path admits its constant-speed reparameterization within the class. Define

$$D(x, \mathcal{C}) = \inf_{\gamma \in \mathcal{P}_T(x, \mathcal{C})} \text{Len}(\gamma), \quad \mathcal{E}_T(x, \mathcal{C}) = \inf_{\gamma \in \mathcal{P}_T(x, \mathcal{C})} \frac{1}{2} \int_0^T \|\dot{\gamma}_t\|^2 dt.$$

THEOREM 4.4 (Length-energy gauge for RDGC). Under the preceding reparameterization assumptions,

$$(P3) \quad \mathcal{E}_T(x, \mathcal{C}) = \frac{D(x, \mathcal{C})^2}{2T}.$$

The identity holds with value $+\infty$. When the common value is finite, a finite-energy path attains the energy minimum if and only if it has constant speed and attains the length minimum. Consequently, whenever the RDGC path class has this reparameterization property,

$$\mathcal{E}_{K, \mathfrak{F}, T}(S_0; H) = \frac{D_{K, \mathfrak{F}}(S_0; H)^2}{2T}.$$

Thus fixed time is a gauge choice that removes the reparameterization degeneracy of the length functional without changing its optimal geometric path.

Remark 4.5 (A checkable intervention-cube margin). The convexity hypothesis (MJ2) follows from a uniform Hessian margin. For example, suppose $\mathcal{U}_0 \subset (-\delta, 1 + \delta)^m$,

$$\text{Hess } W_0 \succeq \alpha_W g, \quad \|\text{Hess } w_i\|_{\text{op}} \leq b_i,$$

uniformly on $[0, T] \times \mathcal{N}$, and $\alpha_W > (1 + \delta) \sum_i b_i$. Then every W_u is geodesically convex on $\mathcal{U}_0$. The analogous condition $\alpha_\Phi > (1 + \delta) \sum_i c_i$ for $\text{Hess } \Phi_0 \succeq \alpha_\Phi g$ and $\|\text{Hess } \phi_i\|_{\text{op}} \leq c_i$ treats the terminal potential. Affine interventions have $b_i = c_i = 0$, as in [Theorem 5.2](#).

4.3. Existence and Hamilton–Jacobi characterization. We next give one sufficient analytic realization. Let $(\mathcal{M}, g)$ be a finite-dimensional Hadamard manifold and $\mathfrak{F} \subset \mathcal{M}$ a nonempty closed geodesically convex set. Admissible paths are absolutely continuous curves in $\mathfrak{F}$ satisfying $\dot{\gamma}(t) \in \mathcal{A}_t(\gamma(t))$ almost everywhere, and

$$\mathcal{A}_{s,t}(\gamma) = \int_s^t \{L_r(\gamma(r), \dot{\gamma}(r)) + U(r, \gamma(r))\} dr.$$

ASSUMPTION 4.6 (Tonelli–viability conditions). *The admissible velocity multifunction has nonempty closed convex values and a Borel graph. In every smooth coordinate chart, the extended integrand*

$$\bar{L}(t, G, Z) = L_t(G, Z) + U(t, G) + \iota_{\mathcal{A}_t(G)}(Z)$$

is a normal integrand: it is Borel measurable in t , lower semicontinuous in (G, Z) for almost every t , and convex in Z . For some $p > 1$, $a > 0$, and $b \geq 0$, uniformly in (t, G, Z) ,

$$\bar{L}(t, G, Z) \geq a \|Z\|_G^p - b.$$

The viability constraint is sequentially closed under uniformly convergent base paths and weakly convergent chart velocities. The terminal cost Γ is proper, lower semicontinuous, and bounded below, and at least one finite-action competitor exists whenever the value under consideration is finite. The path system is composable in the sense of [Theorem 4.1](#).

THEOREM 4.7 (Tonelli existence for restricted paths). *Under [Theorem 4.6](#), every finite value $V(s, x)$ is attained. If pointwise geodesic interpolation preserves admissibility and the complete action plus terminal cost is strictly geodesically convex for paths with a common initial state, then the minimizing path is unique.*

For the HJB layer we use a separate, directly checkable control realization. Suppose $\mathfrak{F}$ is a smooth complete manifold without boundary, $\mathcal{U}$ is a compact metric control space, and admissible paths solve

$$\dot{G}_t = b(t, G_t, a_t)$$

for measurable controls $a_t \in \mathcal{U}$. Assume b is continuous, locally Lipschitz in G uniformly in (t, a) , and tangent to $\mathfrak{F}$. Let the running cost $\ell(t, G, a)$ and terminal cost $\Gamma(G)$ be continuous and bounded below, and define

$$\mathcal{H}(t, G, p) = \sup_{a \in \mathcal{U}} \{ \langle p, b(t, G, a) \rangle - \ell(t, G, a) \}.$$

PROPOSITION 4.8 (HJB layer with explicit analytic boundary). *Assume the preceding compact-control realization is nonexplosive on $[0, T]$, its value V is finite and continuous, and admissible controls are closed under concatenation. Then V is a viscosity solution of*

$$(P4) \quad -\partial_t V(t, G) + \mathcal{H}(t, G, -d_G V(t, G)) = 0, \quad V(T, \cdot) = \Gamma.$$

At every differentiability point this equation holds classically. If the Hamiltonian and terminal data satisfy a comparison principle in a declared solution class, then V is the unique viscosity solution in that class. When the maximizing control is unique, the optimal path satisfies

$$\dot{G}_t = b(t, G_t, a^*(t, G_t, -d_G V(t, G_t)))$$

at classical points. Compactness of $\mathcal{U}$ gives maximizers and local velocity compactness; continuity and the uniform local Lipschitz condition give the short-time realizability used in the viscosity test. No comparison or uniqueness claim is made without the stated comparison hypothesis.

The layers above have different force. Theorem 4.2 is an exact algebraic identity, Theorem 4.4 fixes the time gauge without changing the optimal geometric path, Theorem 4.7 is a direct-method theorem, and Proposition 4.8 is an analytic characterization for the declared compact-control realization, conditional on comparison only for uniqueness.

4.4. Jacobi–Schur response. Let $\mathcal{Z}$ be a finite-dimensional visible parameter space and $\mathcal{H}_0$ a real Hilbert space of zero-boundary path fluctuations. A local trivialization of the admissible path space writes

$$\gamma = Rz + \eta, \quad z \in \mathcal{Z}, \quad \eta \in \mathcal{H}_0,$$

where R is a fixed continuous right inverse for the visible boundary or constraint map. Let $\mathfrak{A} : \mathcal{Z} \times \mathcal{H}_0 \rightarrow \mathbb{R}$ be the resulting action. We use the Riesz isomorphism to identify $\mathcal{H}_0^*$ with $\mathcal{H}_0$.

THEOREM 4.9 (Jacobi–Schur path-response law). Suppose $\mathfrak{A} \in C^3$ near (z_0, η_0) , where η_0 is the unique local minimizer at z_0 . Assume the vertical second variation

$$\mathcal{J}_{\text{Jac}} = D_{\eta\eta}^2 \mathfrak{A}(z_0, \eta_0)$$

is represented by a bounded self-adjoint operator satisfying

$$\langle h, \mathcal{J}_{\text{Jac}} h \rangle \geq \mu \|h\|^2, \quad \mu > 0, \quad h \in \mathcal{H}_0.$$

Then there are neighborhoods Z_0 of z_0 and N_0 of η_0 such that every $z \in Z_0$ has a unique minimizing critical point $\eta_*(z) \in N_0$, and this local minimizing section is C^2 . Write

$$\mathcal{J}_{\text{Jac}}(z) = D_{\eta\eta}^2 \mathfrak{A}(z, \eta_*(z)).$$

With $\text{Val}(z) = \mathfrak{A}(z, \eta_*(z))$,

$$(P5) \quad \boxed{D\eta_* = -\mathcal{J}_{\text{Jac}}^{-1} \mathfrak{A}_{\eta z}},$$

and

$$(P6) \quad \boxed{D^2 \text{Val} = \mathfrak{A}_{zz} - \mathfrak{A}_{z\eta} \mathcal{J}_{\text{Jac}}^{-1} \mathfrak{A}_{\eta z}}.$$

All derivatives on the right are evaluated along the minimizing path. The effective value curvature is therefore the Schur complement of the vertical Jacobi block.

COROLLARY 4.10 (Coercive kinetic bridge on a Hadamard family). *Let $\mathfrak{F}$ be a finite-dimensional Hadamard manifold realized as a totally geodesic submanifold of the ambient geometry, and let γ_0 be the constant-speed geodesic from $x \in \mathfrak{F}$ to a fixed endpoint $q \in \mathfrak{F}$. On zero-boundary fluctuations $h \in H_0^1(\gamma_0^* T\mathfrak{F})$, the second variation of kinetic energy is*

$$I(h, h) = \int_0^T \left(\|D_t h\|^2 - \langle R(\dot{\gamma}_0, h)h, \dot{\gamma}_0 \rangle \right) dt \geq \int_0^T \|D_t h\|^2 dt.$$

Hence the weak Jacobi operator is coercive in the derivative norm on H_0^1 . If q is the unique RDGC target projection, then [Theorem 4.4](#) identifies γ_0 with the energy-gauged RDGC minimizer. Every C^3 endpoint perturbation that preserves the fixed boundary chart therefore satisfies the full Schur formula in [Theorem 4.9](#), including its direct visible Hessian term. Affine additive-action intensities with fixed endpoint additionally satisfy the negative-Gram formula in [Corollary 4.15](#) locally.

4.5. Global Hadamard Möbius–Jacobi realization. The abstract Schur law becomes a global theorem on a whole intervention cube for mechanical actions on Hadamard path space. Let $(\mathcal{N}, g)$ be a finite-dimensional Hadamard manifold, fix $x_0 \in \mathcal{N}$ and $T > 0$, and set

$$\Omega_{x_0} = \{\gamma \in H^1([0, T], \mathcal{N}) : \gamma(0) = x_0\}.$$

Let the open set $\mathcal{U}_0 \Subset \mathcal{U} \subset \mathbb{R}^m$ contain $[0, 1]^m$, and consider

$$W_u = W_0 + \sum_{i=1}^m u_i w_i, \quad \Phi_u = \Phi_0 + \sum_{i=1}^m u_i \phi_i,$$

$$(MJ1) \quad \mathcal{A}_u(\gamma) = \int_0^T \left(\frac{1}{2} \|\dot{\gamma}\|^2 + W_u(t, \gamma(t)) \right) dt + \Phi_u(\gamma(T)).$$

THEOREM 4.11 (Global path-space Möbius–Jacobi theorem). *Fix $r \geq 2$. Assume the bulk potentials are continuous in t , all bulk and terminal potentials are C^{r+2} in the state variable with covariant derivatives locally uniform in (t, u) , and, for every $u \in \mathcal{U}_0$,*

$$(MJ2) \quad \text{Hess } W_u(t, \cdot) \succeq 0, \quad \text{Hess } \Phi_u \succeq 0.$$

Assume also that the values and gradients of $W_u(t, \cdot)$ and Φ_u at x_0 are uniformly bounded over $[0, T] \times \mathcal{U}_0$. Then the following conclusions hold.

1. For every $u \in \mathcal{U}_0$, $\mathcal{A}_u$ has a unique global minimizer γ_u , and $u \mapsto \gamma_u$ is C^r on $\mathcal{U}_0$. No global lower bound on the potentials is required. The minimizer satisfies

$$(MJ3) \quad D_t \dot{\gamma}_u = \text{grad } W_u(t, \gamma_u), \quad \dot{\gamma}_u(T) + \text{grad } \Phi_u(\gamma_u(T)) = 0.$$

2. On $\mathcal{V}_u = \{\xi \in H^1(\gamma_u^* T\mathcal{N}) : \xi(0) = 0\}$, the second variation is

$$(MJ4) \quad \mathfrak{B}_u(\xi, \eta) = \int_0^T [\langle D_t \xi, D_t \eta \rangle - \langle R(\xi, \dot{\gamma}_u) \dot{\gamma}_u, \eta \rangle + \text{Hess } W_u[\xi, \eta]] dt + \text{Hess } \Phi_u[\xi(T), \eta(T)],$$

and it obeys the uniform bound

$$(MJ5) \quad \mathfrak{B}_u(\xi, \xi) \geq \int_0^T \|D_t \xi\|^2 dt.$$

Hence its weak Jacobi map $\mathfrak{J}_u : \mathcal{V}_u \rightarrow \mathcal{V}_u^*$ is a positive self-adjoint isomorphism.

3. Define

$$\Delta_i(\gamma) = \int_0^T w_i(t, \gamma(t)) dt + \phi_i(\gamma(T)), \quad \mathfrak{f}_i^u = D\Delta_i(\gamma_u),$$

and $\chi_i^u = \mathfrak{J}_u^{-1} \mathfrak{f}_i^u$. Then

$$(MJ6) \quad \partial_{u_i} \gamma_u = -\chi_i^u.$$

The response is equivalently the unique solution of

$$(MJ7) \quad \begin{aligned} J_u \chi_i^u &= \text{grad } w_i(t, \gamma_u), & \chi_i^u(0) &= 0, \\ D_t \chi_i^u(T) + \text{Hess } \Phi_u \chi_i^u(T) &= \text{grad } \phi_i(\gamma_u(T)), \end{aligned}$$

where

$$J_u \xi = -D_t^2 \xi - R(\xi, \dot{\gamma}_u) \dot{\gamma}_u + \text{Hess } W_u \xi.$$

Thus bulk and terminal intervention forces share one Green–Jacobi inverse.

4. The value $\mathcal{V}(u) = \mathcal{A}_u(\gamma_u)$ is C^r and

$$(MJ8) \quad \partial_{u_i} \mathcal{V} = \Delta_i(\gamma_u), \quad \partial_{u_i u_j}^2 \mathcal{V} = -\mathfrak{B}_u(\chi_i^u, \chi_j^u).$$

Equivalently, $D_{uu}^2 \mathcal{V} = -\mathfrak{F}_u^* \mathfrak{J}_u^{-1} \mathfrak{F}_u \preceq 0$, where $\mathfrak{F}_u a = \sum_i a_i \mathfrak{f}_i^u$. More precisely,

$$(MJ8a) \quad \begin{aligned} \frac{\|\mathfrak{F}_u a\|_{\mathcal{V}_u^*}^2}{\|\mathfrak{J}_u\|} &\leq -a^\top D_{uu}^2 \mathcal{V}(u) a \\ &\leq \|\mathfrak{J}_u^{-1}\| \|\mathfrak{F}_u a\|_{\mathcal{V}_u^*}^2, \end{aligned}$$

and therefore

$$(MJ8b) \quad \ker(-D_{uu}^2 \mathcal{V}) = \ker \mathfrak{F}_u, \quad \text{rank}(-D_{uu}^2 \mathcal{V}) = \text{rank } \mathfrak{F}_u.$$

5. For $F(A) = \mathcal{V}(\mathbf{1}_A)$ and $\zeta_S = \sum_{B \subseteq S} (-1)^{|S|-|B|} F(B)$, every $|S| \leq r$ satisfies

$$(MJ9) \quad \zeta_S = \int_{[0,1]^S} \partial_S^{|S|} \mathcal{V} \left(\sum_{i \in S} u_i e_i \right) \prod_{i \in S} du_i.$$

In particular,

$$(MJ10) \quad \boxed{\zeta_{\{i,j\}} = - \int_0^1 \int_0^1 \mathfrak{B}_{se_i + te_j}(\chi_i^{se_i + te_j}, \chi_j^{se_i + te_j}) ds dt.}$$

The same formula on the face $u = \mathbf{1}_C + se_i + te_j$ gives the conditional pair contrast for every background $C \cap \{i, j\} = \emptyset$, with every coordinate outside $C \cup \{i, j\}$ fixed at zero.

COROLLARY 4.12 (Arbitrary finite-order Green–Jacobi recursion). *Choose a smooth local path chart, let $I \subseteq [m]$ satisfy $1 \leq |I| \leq r$, and write $\eta_I^u = \partial_I^{|I|} \gamma_u$, and interpret the displayed higher derivatives in that chart (equivalently, under one fixed covariant differentiation convention). Let $\Pi(J)$ denote the set of partitions of J , with $\Pi(\emptyset) = \{\emptyset\}$, and put $\mathcal{F}(u, \gamma) = D_\gamma \mathcal{A}_u(\gamma)$. Then*

$$(MJ11) \quad \eta_I^u = -\mathfrak{J}_u^{-1} \mathcal{R}_I^u,$$

where

$$(MJ12) \quad \mathcal{R}_I^u = \sum_{\substack{S \subseteq I, |S| \leq 1, \pi \in \Pi(I \setminus S) \\ (S, \pi) \neq (\emptyset, \{I\})}} D_u^S D_\gamma^{|\pi|} \mathcal{F}[\eta_B^u]_{B \in \pi}.$$

Thus the same inverse Jacobi operator recursively generates every path derivative through order r and, through (MJ9), every Boolean Möbius effect of that order. Since r may be prescribed arbitrarily, the result covers any fixed finite order under the corresponding C^{r+2} regularity; C^∞ data give all finite orders simultaneously. If $r \geq 3$, then for distinct i, j, k ,

$$(MJ13) \quad \begin{aligned} \partial_{u_i u_j u_k}^3 \mathcal{V} = & -D_\gamma^3 \mathcal{A}_u[\chi_i, \chi_j, \chi_k] \\ & + D_\gamma^2 \Delta_i[\chi_j, \chi_k] + D_\gamma^2 \Delta_j[\chi_i, \chi_k] + D_\gamma^2 \Delta_k[\chi_i, \chi_j]. \end{aligned}$$

The intermediate chart derivatives η_I^u and the displayed recursive representation depend on the chosen path chart, or equivalently on the fixed covariant-differentiation convention. The scalar derivatives $\partial_I^{|I|} \mathcal{V}$ and the resulting Boolean Möbius effects are intrinsic.

4.6. Hard-target response on a smooth active stratum. We now differentiate the terminal constraint itself. This gives the missing hard-target counterpart of the Robin response above. Let $c : \mathcal{U} \times \mathcal{N} \rightarrow \mathbb{R}$ be smooth and consider

$$(HT1) \quad \mathcal{V}_{\text{hard}}(u) = \inf\{\mathcal{A}_u(\gamma) : \gamma \in \Omega_{x_0}, c_u(\gamma(T)) \leq 0\}.$$

The statement is local in the parameter and in one smooth active stratum; the inequality itself remains hard.

THEOREM 4.13 (Bordered Jacobi response for an active hard target). *Fix $u_0 \in \mathcal{U}$. Suppose that $\mathcal{A}_u$ and c_u are C^{r+2} , $r \geq 2$, in compatible local path and state charts. Let γ_0 be a constrained local minimizer of (HT1) with active constraint, and suppose that there is a strictly positive multiplier λ_0 such that, for*

$$\mathcal{L}(u, \gamma, \lambda) = \mathcal{A}_u(\gamma) + \lambda c_u(\gamma(T)),$$

the KKT equations are

$$(HT2) \quad D_\gamma \mathcal{L}(u_0, \gamma_0, \lambda_0) = 0, \quad c_{u_0}(\gamma_0(T)) = 0.$$

On $\mathcal{V}_0 = \{\xi \in H^1(\gamma_0^* TN) : \xi(0) = 0\}$, set

$$\mathfrak{J}_0 = D_{\gamma\gamma}^2 \mathcal{L}, \quad \mathfrak{B}_0 \xi = \text{dc}_{u_0}(\gamma_0(T))[\xi(T)].$$

Assume that $\mathfrak{J}_0 : \mathcal{V}_0 \rightarrow \mathcal{V}_0^*$ is bounded, self-adjoint, and coercive, and that $\mathfrak{B}_0 \neq 0$. Then:

1. there is a neighborhood U_0 of u_0 and a unique C^r branch $u \mapsto (\gamma_u, \lambda_u)$ of active KKT points near (γ_0, λ_0) , with $\lambda_u > 0$; every γ_u is the unique constrained local minimizer in a common path neighborhood;

2. on $\mathcal{Y}_u = \mathcal{V}_u \times \mathbb{R}$, the bordered Jacobi–KKT map

$$(HT3) \quad \mathfrak{K}_u = \begin{pmatrix} \mathfrak{J}_u & \mathfrak{B}_u^* \\ \mathfrak{B}_u & 0 \end{pmatrix} : \mathcal{Y}_u \longrightarrow \mathcal{Y}_u^*$$

is a self-adjoint isomorphism, where $\mathfrak{J}_u = D_{\gamma\gamma}^2 \mathcal{L}$ and $\mathfrak{B}_u \xi = \mathrm{d}c_u(\gamma_u(T))[\xi(T)]$;

3. if

$$\mathfrak{b}_i^u = D_{(\gamma, \lambda)} \partial_{u_i} \mathcal{L}(u, \gamma_u, \lambda_u) \in \mathcal{Y}_u^*,$$

then the path and multiplier response and the constrained value Hessian are

$$(HT4) \quad \partial_{u_i}(\gamma_u, \lambda_u) = -\mathfrak{K}_u^{-1} \mathfrak{b}_i^u,$$

$$(HT5) \quad \partial_{u_i u_j}^2 \mathcal{V}_{\text{hard}} = \partial_{u_i u_j}^2 \mathcal{L} - \langle \mathfrak{b}_i^u, \mathfrak{K}_u^{-1} \mathfrak{b}_j^u \rangle_{\mathcal{Y}_u^*, \mathcal{Y}_u}.$$

The direct derivative in (HT5) holds (γ, λ) fixed.

If $\mathcal{A}_u$ and c_u are affine in u , the direct term in (HT5) vanishes. If the same active branch and bordered invertibility persist on a Boolean two-face, its exact conditional pair effect is

$$(HT6) \quad \zeta_{\{i, j\}|C} = - \int_0^1 \int_0^1 \langle \mathfrak{b}_i^u, \mathfrak{K}_u^{-1} \mathfrak{b}_j^u \rangle \, \mathrm{d}s \, \mathrm{d}t, \quad u = \mathbf{1}_C + se_i + te_j.$$

Here every coordinate outside $C \cup \{i, j\}$ is fixed at zero. Unlike the unconstrained Gram formula, the bordered inverse is indefinite, so hard-target interactions need not have one sign.

The terminal contribution to $\mathfrak{J}_u$ is $\lambda_u \text{Hess } c_u$, while $\mathfrak{B}_u$ linearizes motion of the active boundary. Thus (HT3) couples the interior Jacobi equation, endpoint transversality, and active-target motion in one invertible system.

For fixed endpoints, the same theorem holds on Dirichlet fluctuations and the terminal forces disappear. For a general regular Lagrangian it remains valid when existence, uniqueness, smoothness, and uniform coercivity of the complete second variation are verified. At active-set changes the correct object is a normal-cone graphical derivative coupled to a Jacobi variational inequality; a classical Hessian need not exist.

Accordingly, [Theorem 4.11](#) is the smooth mechanical extension of the RDGC path framework, while [Theorem 4.13](#) and [Corollary 3.7](#) gives the classical hard-target response on a fixed smooth active stratum. Neither theorem asserts classical differentiability through eigenvalue multiplicities or active-index changes.

The theorem applies to a smooth integral action when its second-variation bilinear form is continuous and coercive on the chosen Sobolev fluctuation space. The Riesz map identifies that bilinear form with $\mathcal{J}_{\text{Jac}}$; in a classical Euler–Lagrange problem this is the weak Jacobi operator with the declared boundary conditions.

COROLLARY 4.14 (Associative second-order path elimination). *If $\eta = (\eta_1, \eta_2)$ and the complete vertical Hessian is coercive, then eliminating η_2 and subsequently η_1 gives the same value Hessian as eliminating (η_1, η_2) in one step:*

$$(P7) \quad \text{Schur}_{(\eta_1, \eta_2)} \mathbb{H} = \text{Schur}_{\eta_1} (\text{Schur}_{\eta_2} \mathbb{H}).$$

4.7. Interaction curvature generated by path relaxation. Let $u \in U \subset \mathbb{R}^m$ parameterize smooth mechanism intensities and suppose the unreduced action is affine in u :

$$\mathfrak{A}(\eta, u) = \mathfrak{A}_0(\eta) + \sum_{i=1}^m u_i \Delta_i(\eta).$$

Define $\mathcal{G} : \mathbb{R}^m \rightarrow \mathcal{H}_0^*$ by

$$\mathcal{G}a = \sum_i a_i D_\eta \Delta_i.$$

The finite-dimensional form of the following Schur identity is established in the companion hidden-state reduction theory [30]. We record its path-Hilbert form to connect the abstract reduction formula to the Jacobi and Green operators constructed above; the generic negative-Gram algebra is used as an established ingredient.

COROLLARY 4.15 (Path interaction curvature). *Under Theorem 4.9, the local reduced value satisfies*

$$(P8) \quad \boxed{D_{uu}^2 \text{Val} = -\mathcal{G}^* \mathcal{J}_{\text{Jac}}^{-1} \mathcal{G} \preceq 0.}$$

In components,

$$(P9) \quad \partial_{u_i u_j}^2 \text{Val} = -\langle D_\eta \Delta_i, \mathcal{J}_{\text{Jac}}^{-1} D_\eta \Delta_j \rangle.$$

If the same minimizing branch exists smoothly and the vertical second variation remains uniformly coercive on an open set containing $[0, 1]^m$, then for two Boolean endpoint settings, with all other intensities fixed at zero, the mixed finite difference is exactly

$$(P10) \quad \begin{aligned} & \text{Val}(e_i + e_j) - \text{Val}(e_i) - \text{Val}(e_j) + \text{Val}(0) \\ &= - \int_0^1 \int_0^1 \langle D_\eta \Delta_i, \mathcal{J}_{\text{Jac}}^{-1} D_\eta \Delta_j \rangle_{\eta_*(se_i + te_j)} \, ds \, dt. \end{aligned}$$

Thus hidden path relaxation creates an observable interaction whenever the two path perturbations fail to be orthogonal in the inverse-Jacobi metric. A zero integrated interaction can also arise through cancellation; only pointwise zero curvature implies pointwise orthogonality.

5. A Closed-Form Diagonal RDGC Model. The general theory has a non-trivial realization in the determinant-one two-dimensional diagonal family. Write

$$D(u) = \text{diag}(e^{u/2}, e^{-u/2}),$$

so the global scale has been quotiented out. The induced affine-invariant length is

$$\text{Len}_{\text{diag}}(u_{[0,T]}) = \frac{1}{\sqrt{2}} \int_0^T |\dot{u}_t| \, dt.$$

For $H \in \mathbb{S}_{++}^2$, define

$$K_{\text{diag}}^*(H) = \inf_{u \in \mathbb{R}} \kappa_{\text{gen}}(H, D(u)).$$

THEOREM 5.1 (Two-dimensional diagonal exact complexity). *Let $H \in \mathbb{S}_{++}^2$, $K \geq 1$, and*

$$R_K = \frac{K+1}{\sqrt{K}} \sqrt{\det H}.$$

If $K < K_{\text{diag}}^*(H)$, then

$$D_{K,\text{diag}}^{2D}(u_0; H) = +\infty.$$

If $K \geq K_{\text{diag}}^*(H)$, define

$$x_{\pm}(K) = \frac{R_K \pm \sqrt{R_K^2 - 4H_{11}H_{22}}}{2H_{22}}, \quad u_{\pm}(K) = 2 \log x_{\pm}(K).$$

Then the feasible set is

$$\mathcal{I}_K(H) = [u_-(K), u_+(K)]$$

and

$$D_{K,\text{diag}}^{2D}(u_0; H) = \frac{1}{\sqrt{2}} \text{dist}(u_0, \mathcal{I}_K(H)).$$

The interval $\mathcal{I}_K(H)$ records the exact expressive range of the remaining log-scale ratio. The off-diagonal entry of (H) enters through $\det H$, so correlation shrinks the feasible interval and raises the smallest reachable condition number.

Let

$$q = P_{\mathcal{I}_K(H)}(u_0)$$

be the unique RDGC endpoint and fix $T > 0$. For mechanism intensities $a = (a_1, a_2) \in \mathbb{R}^2$, consider paths $v \in H^1([0, T])$ satisfying $v(0) = u_0$, $v(T) = q$, with action

$$(D1) \quad \mathfrak{A}(v; a) = \frac{1}{4} \int_0^T \dot{v}(t)^2 dt - a_1 \int_0^T v(t) dt - a_2 \int_0^T \frac{t}{T} v(t) dt.$$

THEOREM 5.2 (Exact forced diagonal RDGC response). *Assume $K \geq K_{\text{diag}}^*(H)$. The action in (D1) has the unique minimizer*

$$v_a(t) = u_0 + \frac{q - u_0}{T}t + a_1 w_1(t) + a_2 w_2(t),$$

where

$$w_1(t) = t(T - t), \quad w_2(t) = \frac{Tt}{3} - \frac{t^3}{3T}.$$

At $a = 0$, this is the constant-speed RDGC path and

$$\mathfrak{A}(v_0; 0) = \frac{D_{K,\text{diag}}^{2D}(u_0; H)^2}{2T}.$$

For $\text{Val}(a) = \min_v \mathfrak{A}(v; a)$, the path response is $\partial_{a_i} v_a = w_i$, and

$$(D2) \quad D_a^2 \text{Val} = -T^3 \begin{pmatrix} 1/6 & 1/12 \\ 1/12 & 2/45 \end{pmatrix} \prec 0.$$

The positive matrix inside the minus sign has determinant $1/2160$.

The preceding examples use the unrestricted path class inside a geodesically convex diagonal family. A coordinate-update protocol gives a simple case in which the admissible path geometry cannot be replaced by the ambient AIRM projection in **Proposition 3.3**. Define

$$D_3(x) = \text{diag}(e^{x_1}, e^{x_2}, e^{-x_1 - x_2}), \quad x \in \mathbb{R}^2,$$

and call a path sequential if at most one of $\dot{x}_1, \dot{x}_2$ is nonzero almost everywhere. This models a diagonal preconditioner that rescales one independent log-coordinate at a time.

601 PROPOSITION 5.3 (Exact coordinate-sequential RDGC). For $H = I$, put $k =$
 602 $\log K$ and

$$603 \quad \mathcal{H}_k = \{y \in \mathbb{R}^2 : \max(y_1, y_2, -y_1 - y_2) - \min(y_1, y_2, -y_1 - y_2) \leq k\}.$$

604 The sequential path distance and RDGC are

$$605 \quad (D7) \quad d_{\text{seq}}(x, y) = \sqrt{2} \|x - y\|_1, \quad D_{K, \text{seq}}^{3D}(x_0; I) = \sqrt{2} \min_{y \in \mathcal{H}_k} \|x_0 - y\|_1.$$

606 In particular, if $x_0 = (a, a)$ with $a > k/3$, the unique symmetric target point is
 607 $q = (k/3, k/3)$, and

$$608 \quad (D8) \quad D_{K, \text{seq}}^{3D}(x_0; I) = 2\sqrt{2} \left(a - \frac{k}{3}\right),$$

$$d_{\text{AI}}(D_3(x_0), D_3(q)) = \sqrt{6} \left(a - \frac{k}{3}\right).$$

609 Thus the sequential update protocol has a strictly larger exact RDGC, by the factor
 610 $2/\sqrt{3}$, than the ambient AIRM projection benchmark.

611 The fixed-endpoint model verifies the unconstrained Green response directly and
 612 shows that two time profiles acquire a nonzero interaction solely through relaxation
 613 of a structured metric path. The next model keeps the terminal condition hard and
 614 moves its active spectral boundary.

615 Let $b \in \mathbb{R}^m$, put $\rho(u) = \rho_0 + b^\top u$, and define the determinant-one Hessian family

$$616 \quad H_u = \text{diag}(e^{\rho(u)/2}, e^{-\rho(u)/2}).$$

617 THEOREM 5.4 (Exact moving hard-target diagonal response). Let $K > 1$, and
 618 suppose an open set $\mathcal{U}_0 \supset [0, 1]^m$ satisfies

$$619 \quad (D3) \quad v_0 > \rho(u) + \log K, \quad u \in \mathcal{U}_0.$$

620 For the metric path $D(v(t))$, impose the hard terminal condition

$$621 \quad \kappa_{\text{gen}}(H_u, D(v(T))) \leq K$$

622 and minimize the kinetic action $\frac{1}{4} \int_0^T \dot{v}^2 dt$. Then the active target endpoint, path,
 623 multiplier, and value are

$$624 \quad (D4) \quad q_u = \rho(u) + \log K,$$

$$v_u(t) = v_0 + \frac{q_u - v_0}{T} t,$$

$$\lambda_u = \frac{v_0 - q_u}{2T} > 0,$$

$$\mathcal{V}_{\text{hard}}(u) = \frac{(v_0 - q_u)^2}{4T}.$$

625 Consequently,

$$626 \quad (D5) \quad \partial_{u_i} v_u(t) = b_i \frac{t}{T}, \quad \partial_{u_i} \lambda_u = -\frac{b_i}{2T}, \quad \boxed{D_{uu}^2 \mathcal{V}_{\text{hard}} = \frac{bb^\top}{2T}}.$$

Every Boolean conditional pair effect on the cube is therefore

$$(D6) \quad \zeta_{\{i,j\}|C}^{\text{hard}} = \frac{b_i b_j}{2T}.$$

Thus equally oriented target shifts have positive hard interaction and oppositely oriented shifts have negative hard interaction. The whole cube remains on one simple active spectral stratum by (D3), so (D4)–(D6) instantiate (HT4)–(HT6).

6. Deterministic Verification. The source tree contains a finite-dimensional verification of identities that are algebraically independent of the proofs. A nodal discretization of a strictly convex quartic path action is solved under two affine bulk/terminal interventions. The exact tridiagonal Hessian supplies every Jacobi solve, and tensor Gauss–Legendre quadrature integrates the curvature over the intervention square.

At $N = 40$ nodes and quadrature order $q = 10$, the four-vertex effect is

$$\zeta_{\{1,2\}} = -0.15744124517954516, \quad I_{\text{Jac}} = -0.15744124517876962.$$

Their absolute discrepancy is 7.755×10^{-13} , and the maximum Green-reciprocity error is 8.327×10^{-17} . The minimum sampled coordinate-Hessian eigenvalue is 0.1041677, which certifies positive definiteness of this discrete problem only; its value is basis- and mesh-scaled.

The same program verifies the diagonal interaction matrix and determinant in (D2), both signs of the moving hard-target interaction in (D6), and the sequential-to-ambient ratio in (D8). A separate symbolic check verifies the scalar implicit-differentiation specialization of the third-order formula; it is not a numerical validation of the full path-space recursion in [Corollary 4.12](#).

The deterministic entry point is `experiments/toy/verify_mobius_jacobi.py`; passing `--sensitivity` produces [Table 1](#). No random seed is used. The reported run used Python 3.11.4, NumPy 1.26.4, SciPy 1.10.1, and SymPy 1.11.1. These checks compare two evaluations of the same discretized identities. They establish discrete algebraic consistency, with no continuous-discretization error or optimizer-performance claim.

7. Discussion. The results separate four logical layers of restricted metric optimization. Path concatenation gives the exact min-plus semigroup. Tonelli assumptions give minimizers. A compact-control representation supplies the conditional Hamilton–Jacobi layer. Smooth nondegeneracy gives the Jacobi inverse and its Schur-complement response. In the Hadamard mechanical model, nonpositive curvature and convex potentials derive that nondegeneracy globally on an entire neighborhood of the intervention cube.

The fixed-horizon energy gauge is essential. It removes the reparameterization zero mode of length while preserving the RDGC minimizer and fixes the action value at $D_{K,\mathfrak{F}}^2/(2T)$. The resulting Green operator handles bulk and terminal Robin forcing in one system, generates the value Hessian with exact spectral and rank information, and recursively determines every prescribed finite interaction order under corresponding regularity. When the condition target itself moves, the bordered Jacobi–KKT operator adds the active boundary and its multiplier to the same response calculation. This supplies a classical hard-RDGC response on each regular active spectral stratum.

The strongest statements are finite-dimensional and deterministic. A nonlinear training problem requires a separate realization theorem connecting optimizer state, damping, bias correction, and interpolation to an admissible metric path. Eigenvalue

TABLE 1

Deterministic mesh and quadrature sensitivity. The effect changes with the mesh, while the two discrete evaluations agree to approximately machine precision. The coordinate eigenvalue is not a discretization of a mesh-independent coercivity constant.

N	four-vertex effect	identity error	coordinate $\lambda_{\min}$
10	$-1.7647773 \times 10^{-1}$	9.398×10^{-13}	3.963001×10^{-1}
20	$-1.6369020 \times 10^{-1}$	1.123×10^{-12}	2.049008×10^{-1}
40	$-1.5744125 \times 10^{-1}$	7.755×10^{-13}	1.041677×10^{-1}
80	$-1.5435403 \times 10^{-1}$	4.931×10^{-13}	5.251650×10^{-2}
q at $N = 40$	4	6	10 / 16
identity error	1.008×10^{-10}	1.267×10^{-12}	$7.755/7.507 \times 10^{-13}$

collisions, nonsmooth active-set changes, rank changes, multiple minimizers, and loss of Jacobi coercivity require stratified or generalized response theory. These boundaries delimit the classical derivatives proved here.

In summary, RDGC is a genuine path-space value: its global elimination law is Bellman composition, its smooth local response is a Jacobi Schur complement, and its finite intervention effects are exact integrals of a Green or bordered Green kernel. The closed-form diagonal model demonstrates the unconstrained layers within a restricted preconditioning geometry. Its moving-Hessian specialization additionally gives the hard spectral endpoint, multiplier, and both possible pair-interaction signs in closed form. The coordinate-sequential three-dimensional example shows separately that an update protocol can make RDGC strictly exceed the ambient static projection benchmark.

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Appendix A. Proofs for Path-Space Reduction.

Proof of Theorem 4.2. Fix an admissible path $\gamma \in \mathcal{P}_{s,t}(x, z)$ and put $y = \gamma(r)$. Restriction and action additivity give

$$\mathcal{A}_{s,t}(\gamma) \geq c_{s,r}(x, y) + c_{r,t}(y, z).$$

Taking the infimum over γ proves that the left side of (P1) is at least the right side.

If the right side is $+\infty$, the preceding inequality already forces the left side to be $+\infty$. Otherwise, for every $\varepsilon > 0$, choose y whose two finite transition costs have sum within ε of the right side, and choose segments γ_1, γ_2 whose actions are within ε of those costs. Their concatenation is admissible and has action at most

$$\inf_w \{c_{s,r}(x, w) + c_{r,t}(w, z)\} + 3\varepsilon.$$

Let $\varepsilon \downarrow 0$. The interval lower bounds in Theorem 4.1 exclude $-\infty$ transition costs, so every sum used above is defined. $\square$

Proof of Theorem 4.11. We divide the argument into the global variational, differential, and Boolean layers.

Path norm and action regularity.. For $\xi \in \mathcal{V}_u$, parallel transport to one tangent space and the fundamental theorem of calculus give

$$(A.P1) \quad \|\xi(T)\|^2 \leq T \int_0^T \|D_t \xi\|^2 dt, \quad \int_0^T \|\xi(t)\|^2 dt \leq \frac{T^2}{2} \int_0^T \|D_t \xi\|^2 dt.$$

Hence the derivative norm is equivalent to the usual H^1 norm on the fixed-initial fluctuation space. The hypotheses of [Lemma A.1](#) hold with $k = r + 2$; consequently $(u, \gamma) \mapsto \mathcal{A}_u(\gamma)$ is a C^{r+1} functional in every exponential path chart used below.

Coercivity and existence without a potential lower bound.. Geodesic convexity gives the global support inequalities

$$W_u(t, x) \geq W_u(t, x_0) + \langle \text{grad } W_u(t, x_0), \log_{x_0} x \rangle,$$

$$\Phi_u(x) \geq \Phi_u(x_0) + \langle \text{grad } \Phi_u(x_0), \log_{x_0} x \rangle.$$

The uniform base-point bounds imply that both right sides are bounded below by $-C(1 + d(x_0, x))$, uniformly in (t, u) . Put $E(\gamma) = \int_0^T \|\dot{\gamma}\|^2 dt$. Since

$$d(x_0, \gamma(t)) \leq \sqrt{t} E(\gamma)^{1/2},$$

there are constants C_1, C_2 , independent of $u \in \mathcal{U}_0$, such that

$$(A.P2) \quad \mathcal{A}_u(\gamma) \geq \frac{1}{2} E(\gamma) - C_1 E(\gamma)^{1/2} - C_2.$$

Thus minimizing sequences have uniformly bounded H^1 energy even when a potential tends linearly to $-\infty$.

The common initial point and the bound

$$d(\gamma(s), \gamma(t)) \leq |t - s|^{1/2} E(\gamma)^{1/2}$$

confine a minimizing sequence to a common bounded ball and make it equicontinuous. Finite-dimensional Hadamard manifolds are proper, so Arzela–Ascoli gives a uniformly convergent subsequence. In a finite family of path charts its velocities converge weakly in L^2 . Kinetic energy is weakly lower semicontinuous, while uniform path convergence and local uniform continuity of the potentials give convergence of the bulk and terminal potential terms. The limit therefore attains the infimum.

Strict path convexity and uniqueness.. For two admissible paths γ^0, γ^1 , let $\Gamma(s, t)$ be their pointwise geodesic homotopy and write $H = \partial_s \Gamma$, $V = \partial_t \Gamma$. The common initial point gives $H(s, 0) = 0$. The second variation along the homotopy is

$$\begin{aligned} \frac{d^2}{ds^2} \mathcal{A}_u(\Gamma_s) &= \int_0^T \left[\|D_t H\|^2 - \langle R(H, V)V, H \rangle + \text{Hess } W_u[H, H] \right] dt \\ &\quad + \text{Hess } \Phi_u[H(T), H(T)]. \end{aligned}$$

Every term after $\|D_t H\|^2$ is nonnegative by nonpositive curvature and (MJ2). If the lower bound vanishes, then $D_t H = 0$; together with $H(s, 0) = 0$, this implies $H \equiv 0$. Hence the action is strictly convex along every nontrivial path homotopy. The minimizer is unique.

Euler–Lagrange equation and index form.. For $\xi(0) = 0$, the first variation at the minimizer is

$$\begin{aligned} D\mathcal{A}_u(\gamma_u)[\xi] &= \int_0^T \langle -D_t \dot{\gamma}_u + \text{grad } W_u, \xi \rangle dt \\ &\quad + \langle \dot{\gamma}_u(T) + \text{grad } \Phi_u, \xi(T) \rangle. \end{aligned}$$

Compactly supported tests and then arbitrary terminal values give (MJ3) in weak form. In a coordinate chart the weak equation first gives $\dot{\gamma}_u \in W^{1,1}$, hence $\gamma_u \in C^1$;

continuity in t of the force and smoothness of the connection then give $\gamma_u \in C^2$ and (MJ3) pointwise. Differentiating once more gives (MJ4). Nonpositive curvature and (MJ2) give (MJ5) term by term. The inequalities in (A.P1) show that $\mathfrak{B}_u$ is continuous and coercive on $\mathcal{V}_u$. Lax–Milgram therefore makes $\mathfrak{J}_u : \mathcal{V}_u \rightarrow \mathcal{V}_u^*$ an isomorphism, with inverse norm at most one in the derivative norm.

Smooth global branch.. In a path chart, write the first-order condition as

$$\mathcal{F}(u, \gamma) = D_\gamma \mathcal{A}_u(\gamma) = 0.$$

Its path derivative at γ_u is $\mathfrak{J}_u$, hence invertible. The Hilbert implicit-function theorem produces a local C^r critical branch around every $u \in \mathcal{U}_0$. Strict convexity makes every critical path the unique global minimizer, so local branches agree on overlaps and form one C^r map on the whole cube neighborhood.

Weak and strong Green–Jacobi response.. Differentiating $\mathcal{F}(u, \gamma_u) = 0$ in coordinate u_i gives

$$\mathfrak{J}_u(\partial_{u_i} \gamma_u) + \mathfrak{f}_i^u = 0,$$

which proves (MJ6). For smooth ξ, η , covariant integration by parts in (MJ4) gives

$$\begin{aligned} \mathfrak{B}_u(\xi, \eta) &= \int_0^T \langle J_u \xi, \eta \rangle dt \\ &\quad + \langle D_t \xi(T) + \text{Hess } \Phi_u \xi(T), \eta(T) \rangle. \end{aligned}$$

The closed coercive form corresponds to the Friedrichs self-adjoint operator with homogeneous conditions $\xi(0) = 0$ and $D_t \xi(T) + \text{Hess } \Phi_u \xi(T) = 0$. Comparing $\mathfrak{B}_u(\chi_i^u, \eta) = \mathfrak{f}_i^u[\eta]$ with (A.P3), first for compactly supported tests and then for arbitrary endpoint traces, yields the bulk equation and nonhomogeneous Robin condition in (MJ7). One-dimensional elliptic regularity gives the unique classical response. This also identifies the weak inverse, Friedrichs resolvent, and Green response.

Value curvature and reciprocity.. The envelope identity gives

$$\partial_{u_i} \mathcal{V} = \Delta_i(\gamma_u).$$

Differentiating and using (MJ6) yields

$$\partial_{u_i u_j}^2 \mathcal{V} = -\mathfrak{f}_i^u[\chi_j^u] = -\mathfrak{B}_u(\chi_i^u, \chi_j^u).$$

Symmetry of $\mathfrak{B}_u$ proves reciprocity and (MJ8). Applying the identity to a linear combination of interventions gives the negative semidefinite Gram representation. Under the Riesz identification, the spectral bounds

$$\|\mathfrak{J}_u\|^{-1} I \preceq \mathfrak{J}_u^{-1} \preceq \|\mathfrak{J}_u^{-1}\| I$$

give (MJ8a). Since $\mathfrak{J}_u^{-1}$ is strictly positive, $\langle f, \mathfrak{J}_u^{-1} f \rangle = 0$ holds exactly when $f = 0$. This proves the kernel and rank identities in (MJ8b).

Boolean integration.. For a smooth scalar function, the coordinate difference satisfies

$$f(u + e_i) - f(u) = \int_0^1 \partial_{u_i} f(u + s e_i) ds$$

when $u_i = 0$. Iterating this identity over $i \in S$ proves (MJ9). Substitution of (MJ8) proves (MJ10). Starting the same integration from $\mathbf{1}_C$ proves the conditional-face formula. $\square$

LEMMA A.1 (Smoothness of the mechanical action in a path chart). *Let $\bar{\gamma} \in H^1([0, T], \mathcal{N})$, and use a smooth exponential path chart $\gamma = \exp_{\bar{\gamma}} \xi$ on an H^1 -neighborhood of $\bar{\gamma}$. Suppose that the metric coefficients are smooth and that a bulk potential $W_u(t, x)$ is continuous in t , C^k in (u, x) , and has state and parameter derivatives through order k locally uniform in t . Then*

$$(u, \xi) \mapsto \int_0^T \left(\frac{1}{2} \|\dot{\gamma}\|^2 + W_u(t, \gamma(t)) \right) dt$$

is C^{k-1} on a sufficiently small path-chart neighborhood. A C^k terminal potential composed with the endpoint trace has the same regularity. Its derivatives are the variational derivatives obtained by differentiating under the integral and at the endpoint.

Proof. The embedding $H^1([0, T]) \hookrightarrow C^0([0, T])$ is continuous, and one-dimensional H^1 is a Banach algebra. A small chart ball therefore maps into one compact tube around $\bar{\gamma}([0, T])$. On that tube, all coefficient derivatives used below are uniformly bounded. In local coordinates the kinetic integrand is $g_{ab}(\gamma) \dot{\gamma}^a \dot{\gamma}^b / 2$. Repeated differentiation produces finite sums of bounded Nemytskii coefficients multiplied by at most two L^2 velocity factors and by H^1 variations. Hölder's inequality and the algebra estimate make every derivative through order $k-1$ continuous in the H^1 norm. The potential terms follow from the same Nemytskii estimate with no velocity factors. Finally, endpoint evaluation is a continuous linear map from H^1 to a finite-dimensional tangent space, so ordinary finite-dimensional composition treats the terminal term. $\square$

Proof of Theorem 4.13. Work in a smooth path chart around γ_0 and identify nearby fixed-initial fluctuation spaces with one Hilbert space $\mathcal{V}$. The KKT map is

$$\mathcal{F}(u, \gamma, \lambda) = (D_\gamma \mathcal{L}(u, \gamma, \lambda), c_u(\gamma(T))).$$

Its derivative in (γ, λ) at the reference point is precisely $\mathfrak{K}_0$ in (HT3).

We first prove that this derivative is invertible. Coercivity makes $\mathfrak{J}_0 : \mathcal{V} \rightarrow \mathcal{V}^*$ a positive self-adjoint isomorphism. Since $\mathfrak{B}_0 \neq 0$,

$$\sigma_0 = \mathfrak{B}_0 \mathfrak{J}_0^{-1} \mathfrak{B}_0^* > 0.$$

For prescribed $(f, a) \in \mathcal{V}^* \times \mathbb{R}$, the system

$$\mathfrak{J}_0 \xi + \mathfrak{B}_0^* \beta = f, \quad \mathfrak{B}_0 \xi = a$$

has the unique solution

$$\beta = \sigma_0^{-1} (\mathfrak{B}_0 \mathfrak{J}_0^{-1} f - a), \quad \xi = \mathfrak{J}_0^{-1} (f - \mathfrak{B}_0^* \beta).$$

Hence $\mathfrak{K}_0$ is an isomorphism. The Hilbert implicit-function theorem gives the unique C^r KKT branch. Strict positivity of the multiplier and coercivity persist after shrinking the parameter neighborhood.

For a feasible path γ near γ_u , Taylor expansion of the Lagrangian and stationarity give

$$\mathcal{A}_u(\gamma) - \mathcal{A}_u(\gamma_u) = \mathcal{L}(u, \gamma, \lambda_u) - \mathcal{L}(u, \gamma_u, \lambda_u) - \lambda_u c_u(\gamma(T)).$$

The first difference is positive quadratic to leading order by coercivity, and the last term is nonnegative because $\lambda_u > 0$ and $c_u(\gamma(T)) \leq 0$. After one uniform neighborhood reduction, γ_u is therefore the unique constrained local minimizer there.

Differentiating $\mathcal{F}(u, \gamma_u, \lambda_u) = 0$ gives

$$\mathfrak{K}_u \partial_{u_i}(\gamma_u, \lambda_u) + \mathfrak{b}_i^u = 0,$$

which proves (HT4). Because the active constraint vanishes,

$$\mathcal{V}_{\text{hard}}(u) = \mathcal{L}(u, \gamma_u, \lambda_u).$$

The envelope identity and a second differentiation yield

$$\partial_{u_i u_j}^2 \mathcal{V}_{\text{hard}} = \partial_{u_i u_j}^2 \mathcal{L} + \langle \mathfrak{b}_i^u, \partial_{u_j}(\gamma_u, \lambda_u) \rangle.$$

Substitution of (HT4) proves (HT5). Self-adjointness of the bordered map gives reciprocity. If both data are affine in u , the direct second derivative is zero; applying the coordinate fundamental theorem of calculus twice on the indicated face proves (HT6). $\square$

Proof of Corollary 4.12. Apply the set-indexed multivariate Faa di Bruno formula to $\mathcal{F}(u, \gamma_u) = 0$. Since $\mathcal{A}_u$, and hence $\mathcal{F}$, is affine in u , at most one index can act on the explicit u -argument. The resulting expansion is

$$0 = \sum_{\substack{S \subseteq I, |S| \leq 1 \\ \pi \in \Pi(I \setminus S)}} D_u^S D_\gamma^{|\pi|} \mathcal{F}[\eta_B^u]_{B \in \pi}.$$

The unique term containing the highest derivative η_I^u corresponds to $S = \emptyset$, $\pi = \{I\}$, and equals $\mathfrak{J}_u \eta_I^u$. Moving all other terms to the right and applying $\mathfrak{J}_u^{-1}$ proves (MJ11)–(MJ12). Thus every derivative is generated recursively from lower-order derivatives by the same Green inverse.

For completeness, applying the same partition formula to the scalar composite $\mathcal{A}_u(\gamma_u)$ gives

$$(A.P4) \quad \partial_I^{|\pi|} \mathcal{V} = \sum_{\substack{S \subseteq I, |S| \leq 1 \\ \pi \in \Pi(I \setminus S)}} D_u^S D_\gamma^{|\pi|} \mathcal{A}_u[\eta_B^u]_{B \in \pi},$$

where the term $D_\gamma \mathcal{A}_u[\eta_I^u]$ vanishes by stationarity. Equations (A.P4) and (MJ9) give every higher-order Boolean effect.

For three distinct indices, differentiate $\mathfrak{J}_u \partial_{u_j} \gamma_u + D \Delta_j = 0$ in u_k , use the result to eliminate $\partial_{u_j u_k}^2 \gamma_u$ from the second derivative of $\partial_{u_i} \mathcal{V} = \Delta_i(\gamma_u)$, and substitute $\partial_{u_i} \gamma_u = -\chi_i$. Symmetry of $\mathfrak{J}_u$ cancels the second path-response terms and leaves exactly (MJ13). $\square$

Proof of Corollary 3.5. The AIRM cone is Hadamard, and a structured realization satisfying the stated hypotheses has the same intrinsic nonpositive-curvature property. The information potentials supply (MJ2), so Theorem 4.11 and Corollary 4.12 apply directly. The displayed operator is (MJ7) with the AIRM connection and curvature. Its curvature term is nonnegative in the index form, and (MJ10) gives the pair formula. The final scope statement records exactly where the classical Hessian used by the theorem ceases to exist. $\square$

Proof of Theorem 4.4. For every finite-energy path, Cauchy–Schwarz gives

$$\frac{1}{2} \int_0^T \|\dot{\gamma}_t\|^2 dt \geq \frac{1}{2T} \left(\int_0^T \|\dot{\gamma}_t\| dt \right)^2 = \frac{\text{Len}(\gamma)^2}{2T} \geq \frac{D(x, \mathcal{C})^2}{2T}.$$

Thus $\mathcal{E}_T \geq D^2/(2T)$. If $D < +\infty$, choose paths with length converging to D and reparameterize each at constant speed. The reparameterized paths remain admissible and have energy $\text{Len}(\gamma)^2/(2T)$, proving the reverse inequality after taking the limit. If $D = +\infty$, a finite-energy path would have finite length by Cauchy–Schwarz, a contradiction, so both sides are $+\infty$.

Equality in Cauchy–Schwarz holds exactly when the metric speed is constant almost everywhere. Equality in the second inequality holds exactly for a length minimizer. These two equality conditions characterize the energy minimizers and prove the RDGC specialization. $\square$

Proof of Corollary 4.3. Apply Theorem 4.2 to $c_{s,T}(x, z)$, add $\Gamma(z)$, and regroup the two successive infima:

$$\begin{aligned} V(s, x) &= \inf_z \inf_y \{c_{s,r}(x, y) + c_{r,T}(y, z) + \Gamma(z)\} \\ &= \inf_y \{c_{s,r}(x, y) + V(r, y)\}. \end{aligned}$$

Proof of Theorem 4.7. Take a minimizing sequence with uniformly bounded objective. The lower bound on $\bar{L}$ and the lower bound on Γ give a uniform L^p bound on path speeds. Since $p > 1$, Hölder’s inequality gives uniform equicontinuity. The common initial point and the speed bound confine every path to a common closed bounded ball. Finite-dimensional Hadamard manifolds are proper, so Arzelà–Ascoli gives a uniformly convergent subsequence. The corresponding velocities converge weakly after passage to local charts and a further subsequence. Sequential viability keeps the limit admissible. More explicitly, subdivide the compact limiting path into finitely many intervals whose images lie in normal coordinate charts. For all sufficiently large indices the approximating segments lie in the same charts; their coordinate paths converge strongly and uniformly, while their coordinate velocities converge weakly in L^p . On each subinterval the normal-integrand hypothesis and convexity in velocity give weak lower semicontinuity of the integral functional [10, Chapter 3]. Chart-boundary times form a finite measure-zero set, so summing the chartwise inequalities gives the global lower bound. Therefore

$$\mathcal{A}(\gamma) \leq \liminf_n \mathcal{A}(\gamma_n),$$

and lower semicontinuity of Γ treats the endpoint. Hence the limit attains the value.

If two different minimizing paths existed, their pointwise geodesic interpolation would be admissible and strict geodesic convexity would give a strictly smaller objective, a contradiction. $\square$

Proof of Proposition 4.8. The Bellman identity on $[t, t + h]$ gives

$$V(t, G) = \inf_{a[t, t+h]} \left\{ \int_t^{t+h} \ell(s, \gamma_s, a_s) \, ds + V(t+h, \gamma(t+h)) \right\}.$$

Put

$$q_\varphi(s, X, a) = \ell(s, X, a) + d_X \varphi(s, X)[b(s, X, a)].$$

Let a smooth test function φ touch V from above at (t, G) . For every constant control $a \in \mathcal{U}$, the Bellman inequality and the local maximum of $V - \varphi$ give

$$0 \leq \int_t^{t+h} \ell(s, \gamma_s, a) \, ds + \varphi(t+h, \gamma(t+h)) - \varphi(t, G).$$

Uniform local Lipschitz continuity of b gives $\gamma(t+h) = \exp_G(hb(t, G, a) + o(h))$, uniformly in a . Divide by h , let $h \downarrow 0$, and use compactness of $\mathcal{U}$. The inequality holds for every a , hence

$$-\partial_t \varphi(t, G) + \mathcal{H}(t, G, -d_G \varphi(t, G)) \leq 0,$$

which is the viscosity subsolution inequality for the sign convention in (P4).

If φ touches V from below, choose a measurable control whose short-horizon cost is within $o(h)$ of the Bellman infimum. The local minimum of $V - \varphi$ then yields

$$o(h) \geq \int_t^{t+h} \ell(s, \gamma_s, a_s) ds + \varphi(t+h, \gamma(t+h)) - \varphi(t, G).$$

Along this trajectory, the chain rule rewrites the right side as

$$\int_t^{t+h} [\partial_s \varphi(s, \gamma_s) + q_\varphi(s, \gamma_s, a_s)] ds.$$

The trajectory remains uniformly close to G on the shrinking interval. Uniform continuity on the resulting compact neighborhood gives, pointwise in the measurable control,

$$q_\varphi(s, \gamma_s, a_s) \geq \inf_{a \in \mathcal{U}} q_\varphi(t, G, a) - o(1).$$

After division by h , this proves

$$-\partial_t \varphi(t, G) + \mathcal{H}(t, G, -d_G \varphi(t, G)) \geq 0.$$

No closure or convexification of averaged velocity–cost pairs is used. Continuity of V and Γ supplies the terminal condition. At a differentiability point, the two inequalities coincide. The feedback formula uses an attaining maximizer, which exists by compactness. If comparison holds in the declared class, standard comparison gives uniqueness. $\square$

Proof of Theorem 4.9. The first-order condition is

$$D_\eta \mathfrak{A}(z, \eta_\star(z)) = 0.$$

Coercivity and self-adjointness imply that $\mathcal{J}_{\text{Jac}} : \mathcal{H}_0 \rightarrow \mathcal{H}_0$ is invertible; equivalently one may apply Lax–Milgram to the second-variation form. The Hilbert-space implicit-function theorem therefore produces a unique C^2 critical section near (z_0, η_0) . Coercivity persists after shrinking the neighborhood, so this section remains the unique local minimizer. Differentiating the first-order condition gives

$$\mathfrak{A}_{\eta z} + \mathcal{J}_{\text{Jac}} D\eta_\star = 0,$$

which proves (P5).

The envelope identity is

$$D \text{Val} = \mathfrak{A}_z + \mathfrak{A}_\eta D\eta_\star = \mathfrak{A}_z.$$

Differentiating again and substituting (P5) gives

$$D^2 \text{Val} = \mathfrak{A}_{zz} - \mathfrak{A}_{z\eta} \mathcal{J}_{\text{Jac}}^{-1} \mathfrak{A}_{\eta z},$$

which is (P6). $\square$

Proof of Corollary 4.10. The classical second-variation formula for kinetic energy along a geodesic with fixed endpoints is the displayed index form. Since $\mathfrak{F}$ is totally geodesic, its intrinsic connection and curvature are the restrictions used by that formula. Nonpositive sectional curvature gives

$$\langle R(\dot{\gamma}_0, h)h, \dot{\gamma}_0 \rangle \leq 0,$$

and hence $I(h, h) \geq \int_0^T \|D_t h\|^2 dt$. The derivative seminorm is a norm on H_0^1 , equivalent to its standard Sobolev norm by Poincaré's inequality. Thus the Riesz representative of I is bounded, self-adjoint, and coercive. The length-energy identification and the final sensitivity claim now follow from Theorems 4.4 and 4.9. $\square$

Proof of Corollary 4.14. Let $q(\delta z, \delta \eta_1, \delta \eta_2)$ be the quadratic form defined by the complete Hessian. Coercivity gives unique quadratic minimizers. Hence

$$\inf_{\delta \eta_1, \delta \eta_2} q = \inf_{\delta \eta_1} \left(\inf_{\delta \eta_2} q \right).$$

Each quadratic elimination is represented by the corresponding Schur complement. Equality for every δz proves (P7). $\square$

Proof of Corollary 4.15. Affineness in u gives

$$\mathfrak{A}_{uu} = 0, \quad \mathfrak{A}_{\eta u} = \mathcal{G}.$$

The u -block of (P6) is therefore

$$D_{uu}^2 \text{Val} = -\mathcal{G}^* \mathcal{J}_{\text{Jac}}^{-1} \mathcal{G}.$$

For any $a \in \mathbb{R}^m$,

$$\langle a, D_{uu}^2 \text{Val} a \rangle = -\langle \mathcal{G}a, \mathcal{J}_{\text{Jac}}^{-1} \mathcal{G}a \rangle \leq 0,$$

which proves (P8) and (P9). Applying the one-dimensional fundamental theorem of calculus successively in coordinates u_i and u_j gives

$$\text{Val}(e_i + e_j) - \text{Val}(e_i) - \text{Val}(e_j) + \text{Val}(0) = \int_0^1 \int_0^1 \partial_{u_i u_j}^2 \text{Val}(se_i + te_j) ds dt.$$

Substitution of (P9) proves (P10). $\square$

Appendix B. Proofs for the SPD Specialization and Diagonal Model.

Proof of Theorem 3.1. Let $S_1 \in \mathcal{C}_K$, and let $z_1 \leq \dots \leq z_d$ be its ordered log-eigenvalues. The affine-invariant distance satisfies the spectral lower bound

$$d_{\text{AI}}(S_0, S_1) \geq \|y - z\|_2$$

[7]. Since $S_1 \in \mathcal{C}_K$, the vector z has width at most $\log K$, so $z_i \in [c, c + \log K]$ for some c . Therefore

$$d_{\text{AI}}(S_0, S_1)^2 \geq \sum_{i=1}^d \text{dist}(y_i, [c, c + \log K])^2.$$

Conversely, fix c and set

$$z_i(c) = \Pi_{[c, c + \log K]}(y_i).$$

Choose S_1 with the same eigenvectors as S_0 and log-eigenvalues $z_i(c)$. Then $S_1 \in \mathcal{C}_K$, the matrices commute, and $d_{\text{AI}}(S_0, S_1) = \|y - z(c)\|_2$. Minimizing over c proves the formula. $\square$

Proof of Corollary 3.7. Simplicity of the extreme generalized eigenvalues makes $c_u|_{\mathfrak{F}}$ smooth near (u_0, q_0) . For each fixed u , geodesic convexity of the log-condition number makes $\mathcal{T}_u$ geodesically convex. The AIRM projection is therefore unique, and the fixed-horizon kinetic minimizer is the constant-speed geodesic to that projection by Proposition 3.3 and Theorem 4.4.

Since G_0 lies outside the target, the endpoint constraint is active. Regularity of its restricted differential gives LICQ. Projection transversality has the form

$$\dot{\gamma}_{u_0}(T) + \lambda_0 \text{grad}^{\text{AI}}(c_{u_0}|_{\mathfrak{F}})(q_0) = 0.$$

The multiplier is positive: if it vanished, the terminal velocity and hence the constant speed would vanish, contradicting $G_0 \notin \mathcal{T}_{u_0}$.

The second variation of the endpoint Lagrangian is

$$\int_0^T \left(\|D_t \xi\|^2 - \langle R^{\text{AI}}(\xi, \dot{\gamma}) \dot{\gamma}, \xi \rangle \right) dt + \lambda_0 \text{Hess}^{\text{AI}}(c_{u_0}|_{\mathfrak{F}})[\xi(T), \xi(T)].$$

Nonpositive curvature and geodesic convexity make the last two contributions nonnegative, so this form is coercive in the fixed-initial derivative norm. All hypotheses of Theorem 4.13 hold. Its implicit branch is the unique global projection branch because the endpoint objective is strictly convex and $\mathcal{T}_u$ is convex. The value identity follows from the length–energy gauge, and the response and Hessian formulas follow from (HT4)–(HT6). $\square$

Proof of Proposition 3.3. The function $G \mapsto \log \kappa(G)$ is geodesically convex on the SPD cone, so $\mathcal{C}_K$ is closed and geodesically convex. Hence $\mathcal{T}_{K, \mathfrak{F}}(H)$ is closed and geodesically convex. A nonempty closed geodesically convex subset of a finite-dimensional Hadamard manifold has a unique metric projection. The geodesic from S_0 to this projection stays inside $\mathcal{S}_{\mathfrak{F}}(H)$ and has length equal to the metric distance; every admissible curve has length at least the distance between its endpoints. This proves the RDGC identity and uniqueness.

For a remaining horizon $h = T - t$, Cauchy–Schwarz gives

$$\frac{1}{2} \int_t^T \|\dot{\gamma}_s\|^2 ds \geq \frac{\text{Len}(\gamma)^2}{2h} \geq \frac{d_{\text{AI}}(S, \mathcal{T}_{K, \mathfrak{F}}(H))^2}{2h}.$$

The constant-speed geodesic to the projection attains equality. $\square$

Proof of Proposition 3.4. For $A, B \in \mathbb{S}_{++}^d$, $t \in [0, 1]$, and $1 \leq k \leq d$, exterior powers commute with the weighted geometric mean. Monotonicity of that mean yields

$$\prod_{i=1}^k \lambda_i^\downarrow(A \#_t B) \leq \prod_{i=1}^k \lambda_i^\downarrow(A)^{1-t} \lambda_i^\downarrow(B)^t.$$

Equality holds for $k = d$. After taking logarithms,

$$(B.1) \quad \log \lambda(A \#_t B) \prec (1-t) \log \lambda(A) + t \log \lambda(B).$$

Fix $H \succ 0$, let $G_t = G_0 \#_t G_1$, and set $A_i = H^{-1/2} G_i H^{-1/2}$. Congruence is an affine-invariant isometry and preserves geometric means. The relative eigenvalues $G^{-1/2} H G^{-1/2}$ are the reciprocals of those of $H^{-1/2} G H^{-1/2}$. Multiplication by -1 preserves majorization, and a convex permutation-invariant function is Schur convex. Applying (B.1) and ordinary convexity of ϕ gives

$$\Omega_{\phi, H}(G_t) \leq (1-t) \Omega_{\phi, H}(G_0) + t \Omega_{\phi, H}(G_1).$$

Coordinatewise monotonicity followed by convexity of ρ proves the claim for $\mathcal{J}$.
 Closedness follows from continuity of the finite-dimensional convex spectral functions.
 Finally,

$$\log \lambda((cG)^{-1/2}H(cG)^{-1/2}) = \log \lambda(G^{-1/2}HG^{-1/2}) - (\log c)\mathbf{1},$$

which proves scale invariance. $\square$

Proof of Theorem 5.1. For $D(u) = \text{diag}(e^{u/2}, e^{-u/2})$,

$$\log D(u) = \text{diag}(u/2, -u/2),$$

so the induced line element is $ds^2 = \frac{1}{2} du^2$. Let

$$A(u) = D(u)^{-1/2}HD(u)^{-1/2}.$$

Because $\det D(u) = 1$, one has $\det A(u) = \det H$. If the eigenvalues of $A(u)$ are $\mu \leq \nu$, the condition $\nu/\mu \leq K$ with fixed product $\mu\nu = \det H$ is equivalent to

$$\mu + \nu \leq \left(\sqrt{K} + \frac{1}{\sqrt{K}} \right) \sqrt{\det H} = R_K.$$

Moreover,

$$\text{tr } A(u) = H_{11}e^{-u/2} + H_{22}e^{u/2}.$$

With $x = e^{u/2}$, the target condition is

$$H_{22}x^2 - R_Kx + H_{11} \leq 0.$$

Below $K_{\text{diag}}^*(H)$ the feasible set is empty. Otherwise its roots are $x_{\pm}(K)$, so the feasible set in u -coordinates is $[2 \log x_-(K), 2 \log x_+(K)]$. The one-dimensional line element gives the claimed distance. $\square$

Proof of Theorem 5.2. Write every admissible path as

$$v(t) = v_0(t) + \eta(t), \quad v_0(t) = u_0 + \frac{q - u_0}{T}t, \quad \eta \in H_0^1(0, T).$$

The vertical second variation is

$$D_{vv}^2 \mathfrak{A}[h, k] = \frac{1}{2} \int_0^T \dot{h}(t) \dot{k}(t) dt,$$

which is coercive on $H_0^1(0, T)$. The weak Euler–Lagrange equation is

$$\frac{1}{2} \int_0^T \dot{v}_a \dot{h} dt = a_1 \int_0^T h dt + a_2 \int_0^T \frac{t}{T} h dt.$$

The affine path v_0 has zero weak second derivative, while

$$-\frac{1}{2}w_1'' = 1, \quad -\frac{1}{2}w_2'' = \frac{t}{T}, \quad w_i(0) = w_i(T) = 0.$$

Thus the displayed v_a is the unique global minimizer.

At $a = 0$,

$$\mathfrak{A}(v_0; 0) = \frac{(q - u_0)^2}{4T} = \frac{D_{K, \text{diag}}^{2D}(u_0; H)^2}{2T}.$$

1150 The envelope identity gives

$$1151 \quad \partial_{a_i a_j}^2 \text{Val} = - \int_0^T f_i(t) w_j(t) dt, \quad f_1(t) = 1, \quad f_2(t) = t/T.$$

1152 Direct integration yields

$$1153 \quad \int_0^T w_1 dt = \frac{T^3}{6}, \quad \int_0^T w_2 dt = \int_0^T \frac{t}{T} w_1 dt = \frac{T^3}{12},$$

1154 and

$$1155 \quad \int_0^T \frac{t}{T} w_2 dt = \frac{2T^3}{45}.$$

1156 The determinant of the positive matrix in (D2) is $1/2160$, so it is positive definite and
 1157 $D_a^2 \text{Val} \prec 0$. $\square$

1158 *Proof of Theorem 5.4.* For the diagonal Hessian and metric in the theorem,

$$1159 \quad D(v)^{-1/2} H_u D(v)^{-1/2} = \text{diag} \left(e^{(\rho(u)-v)/2}, e^{(v-\rho(u))/2} \right).$$

1160 Hence

$$1161 \quad \log \kappa_{\text{gen}}(H_u, D(v)) = |v - \rho(u)|,$$

1162 and the hard target in the v -coordinate is the interval $[\rho(u) - \log K, \rho(u) + \log K]$.
 1163 Assumption (D3) keeps v_0 strictly above this interval throughout $\mathcal{U}_0$, so its projection
 1164 is the upper endpoint q_u . The unique kinetic minimizer is the straight path in (D4),
 1165 and direct integration gives the displayed value.

1166 On this active stratum use

$$1167 \quad c_u(v) = v - \rho(u) - \log K \leq 0.$$

1168 The endpoint first variation is $\frac{1}{2} \dot{v}_u(T) \xi(T)$. KKT transversality therefore reads

$$1169 \quad \frac{1}{2} \dot{v}_u(T) + \lambda_u = 0,$$

1170 which gives the positive multiplier in (D4). Differentiating the closed forms gives the
 1171 path and multiplier responses in (D5). Since q_u is affine,

$$1172 \quad \partial_{u_i u_j}^2 \mathcal{V}_{\text{hard}} = \frac{b_i b_j}{2T}.$$

1173 This mixed derivative is constant on every Boolean face, so two applications of the
 1174 coordinate fundamental theorem of calculus prove (D6). Finally, c_u is affine in u ,
 1175 its endpoint differential is one, the multiplier stays positive by (D3), and the kinetic
 1176 Jacobi form is coercive. These are exactly the full-face hypotheses of Theorem 4.13
 1177 and Corollary 3.7. $\square$

1178 *Proof of Proposition 5.3.* The AIRM line element in the log-coordinate chart is

$$1179 \quad ds^2 = dx_1^2 + dx_2^2 + (dx_1 + dx_2)^2.$$

1180 Along a sequential path this reduces to $ds = \sqrt{2}|dx_i|$ on every coordinate segment.
 1181 Hence every such path from x to y has length at least

$$1182 \quad \sqrt{2}(|y_1 - x_1| + |y_2 - x_2|).$$

Changing the first coordinate monotonically and then the second attains this bound, proving the transition-distance formula in (D7). The three log-eigenvalues of $D_3(y)$ are $y_1, y_2, -y_1 - y_2$, so its condition target is exactly $\mathcal{H}_k$. Minimization of the transition distance over that target proves the RDGC formula.

For $x_0 = (a, a)$, symmetry and convexity give a symmetric ambient AIRM projection. On the line $y_1 = y_2 = s$, the target condition is $3|s| \leq k$, so the projection is $q = (k/3, k/3)$. The same point minimizes the ℓ_1 distance: for every $y \in \mathcal{H}_k$,

$$2y_1 + y_2 \leq k, \quad y_1 + 2y_2 \leq k,$$

and hence $y_1 + y_2 \leq 2k/3$. Since $|a - y_i| \geq a - y_i$,

$$\|x_0 - y\|_1 \geq 2a - y_1 - y_2 \geq 2\left(a - \frac{k}{3}\right),$$

with equality at q . Finally,

$$d_{\text{AI}}(D_3(x_0), D_3(q))^2 = 2\left(a - \frac{k}{3}\right)^2 + 4\left(a - \frac{k}{3}\right)^2,$$

which gives (D8) and the strict ratio. $\square$